Functions & Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A car travels 120 miles in 3 hours. Find the average speed.

Apply the fundamental distance-rate-time relationship $d = r \cdot t$: $r = \frac{d}{t} = \frac{120 \text{ miles}}{3 \text{ hours}} = \mathbf{40}$ mph ACT Exam Note: Average speed is always Total Distance divided by Total Time. Never average two speeds directly if the travel times are different.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the least common multiple of 6 and 8.

Prime Factorization Method: $6 = 2^1 \times 3^1$ $8 = 2^3$ The Least Common Multiple (LCM) takes the highest power of every prime factor present: $\text{LCM} = 2^3 \times 3^1 = 8 \times 3 = \mathbf{24}$ ACT Tip: List multiples of the larger number until divisible by the smaller: $8, 16, 24$ (divisible by 6).

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Give the equation of the horizontal line through $(0, 5)$.

A horizontal line has a slope of $m = 0$ and every point on it shares the same y-coordinate. Therefore, its general equation is $y = c$. Since it passes through $(0, 5)$, the equation is $\mathbf{y = 5}$. Contrast: A vertical line through the same point would have undefined slope and equation $x = 0$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the distance between $(0, 0)$ and $(3, 4)$.

Apply the Euclidean distance formula derived from the Pythagorean theorem: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(3-0)^2 + (4-0)^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = \mathbf{5}$ ACT Shortcut: Recognize the fundamental $3 - 4 - 5$ Pythagorean triple immediately.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the interior-angle sum of a pentagon.

Use the polygon interior angle sum formula for an n-gon: $S = (n - 2) \times 180^\circ$ A pentagon has $n = 5$ sides: $S = (5 - 2) \times 180^\circ = 3 \times 180^\circ = \mathbf{540}^\circ$ Bonus: If regular, each interior angle measures $\frac{540^\circ}{5} = 108^\circ$.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For what value of x is $\frac{1}{x-2}$ undefined?

A rational expression is undefined wherever its denominator equals zero (division by zero is undefined in real numbers). Set the denominator to zero: $x - 2 = 0 \Rightarrow \mathbf{x = 2}$ At $x = 2$, the graph of $y = \frac{1}{x-2}$ has a vertical asymptote.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has opposite 6 and adjacent 8. Find $\tan\theta$.

Recall the tangent ratio from SOH CAH TOA: $\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{6}{8}$ Reduce the fraction by dividing numerator and denominator by 2: $\tan\theta = \frac{3}{4} = \mathbf{0.75}$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the midpoint of $(2, 4)$ and $(6, 8)$.

Apply the midpoint formula by averaging the coordinates: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = \left(\frac{2+6}{2}, \frac{4+8}{2}\right) = \left(\frac{8}{2}, \frac{12}{2}\right) = \mathbf{(4, 6)}$

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the circumference of a circle with $r = 7$. Take $\pi = \frac{22}{7}$.

Apply the circle circumference formula $C = 2\pi r$: $C = 2 \times \frac{22}{7} \times 7$ Cancel the factor of 7 in the numerator and denominator: $C = 2 \times 22 = \mathbf{44}$

10 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the smallest value of x in the domain of $f(x) = \sqrt{x-2}$? Show your complete algebraic work.

Need $x - 2 \geq 0 \Rightarrow x \geq 2$. Smallest value: 2.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of $y = -f(x)$ is the graph of $y = f(x)$ transformed how? Show your complete algebraic work.

Final Answer: reflected across the x-axis. Negating the output $f(x)$ flips the graph vertically: reflection across the x-axis.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of $y = f(x - 5)$ is the graph of $y = f(x)$ shifted how? Show your complete algebraic work.

Final Answer: 5 units right. $y = f(x - h)$ shifts right when $h > 0$, left when $h < 0$; here shifts 5 units right.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Practice 3.2: What is the domain of the function $f(x) = \frac{\sqrt{x+5}}{x-3}$?

For the radical in the numerator to be defined, the radicand must be non-negative: $x + 5 \geq 0 \Rightarrow x \geq -5$. For the rational expression to be defined, the denominator cannot equal zero: $x - 3 \neq 0 \Rightarrow x \neq 3$. Combining both restrictions gives $x \geq -5$ and $x \neq 3$, or in interval notation $[-5, 3) \cup (3, \infty)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the inverse function $f^{-1}(x)$ of $f(x) = \frac{5x+2}{3}$?

Swap x and y: $x = \frac{5y+2}{3} \Rightarrow 3x - 2 = 5y \Rightarrow f^{-1}(x) = \mathbf{\frac{3x-2}{5}}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

Level 3 · AdvancedIn the hard Module 2, questions 18–22 are typically high-difficulty multi-step problems with parameters k. If a student has 7 minutes remaining for the last 3 questions, which approach optimizes score yield: attempting all 3 partially, or using Desmos regression and choice elimination to guarantee complete accuracy on 2 of them?

Optimal Strategy: On the Digital SAT's Item Response Theory (IRT) scoring model, fully correct answers yield substantially higher raw theta weights than rushed guesses. Allocating 3.5 minutes each to 2 questions using Desmos graphing/regression guarantees 2 correct hard items (+40–60 scaled points), while placing a rapid educated guess on the 3rd question preserves time with zero guessing penalty.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Plane Geometry

On the enhanced ACT Math test (45 questions in 50 minutes), a student allocates 10 minutes to the first 15 questions and 16 minutes to questions 16 through 30. How many seconds per question remain for the final 15 questions, assuming 3 minutes are reserved at the end to check answers?

Time spent so far $= 10 + 16 = 26$ minutes. Leaving 3 minutes for review leaves $50 - 26 - 3 = 21$ minutes for the final 15 questions. Time per question $= \frac{21 \times 60 \text{ seconds}}{15 \text{ questions}} = \frac{1260}{15} = 84$ seconds per question.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Trigonometry

Practice 1.2: If a and b are positive integers such that the greatest common factor of a and b is 6, and the least common multiple of a and b is 72, what is the product $a \times b$?

By the fundamental identity connecting greatest common factor and least common multiple: $\text{GCF}(a, b) \times \text{LCM}(a, b) = a \times b$. Substitute the given values: $a \times b = 6 \times 72 = 432$. The product is 432.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

Why should you test Choice C (or H) first when back-solving numerical answer choices on the ACT?

ACT numerical choices are ordered from least to greatest (or greatest to least). Testing the middle choice tells you whether to move higher or lower, requiring at most 2 tests.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

What is the greatest common factor (GCF) and least common multiple (LCM) of 24 and 36?

$24 = 2^3 \cdot 3$ and $36 = 2^2 \cdot 3^2$. Thus GCF $= 2^2 \cdot 3 =$ **12** and LCM $= 2^3 \cdot 3^2 =$ **72**.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

For what values of x does $|2x - 5| > 9$ hold?

Split into two rays: $2x - 5 > 9 \Rightarrow x > 7$ or $2x - 57$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given the linear function $f(x) = \frac{4x-7}{3}$, what is the equation of the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+7}{4}$ ✓

b. $f^{-1}(x) = \frac{3x-7}{4}$

c. $f^{-1}(x) = \frac{4x+3}{7}$

d. $f^{-1}(x) = \frac{7x+3}{4}$

Set $y = \frac{4x-7}{3} \Rightarrow 3y = 4x - 7 \Rightarrow 4x = 3y + 7 \Rightarrow x = \frac{3y+7}{4}$. Swapping variables gives $f^{-1}(x) = \frac{3x+7}{4}$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $g(x) = \sqrt{2x + 10}$, what is the domain of $g(x)$ in the set of real numbers?

a. $x \geq -5$ ✓

b. $x > -5$

c. $x \geq 0$

d. $x \leq -5$

The expression under the radical must be non-negative: $2x + 10 \geq 0 \Rightarrow 2x \geq -10 \Rightarrow x \geq -5$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = 2x + 5$ and $g(x) = x^2 - 3$. What is the value of $(g \circ f)(1)$?

a. 11

b. 46 ✓

c. 49

d. 52

Evaluate inner function first: $f(1) = 2(1) + 5 = 7$. Then $(g \circ f)(1) = g(7) = 7^2 - 3 = 49 - 3 = 46$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $h(x) = 3x - 1$ and $(h \circ k)(x) = 6x + 8$, which of the following defines $k(x)$?

a. $k(x) = 2x + 3$ ✓

b. $k(x) = 2x + 9$

c. $k(x) = 3x + 4$

d. $k(x) = 2x + 2$

$h(k(x)) = 3k(x) - 1 = 6x + 8 \Rightarrow 3k(x) = 6x + 9 \Rightarrow k(x) = 2x + 3$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the range of the quadratic function $f(x) = -(x - 4)^2 + 9$?

a. $y \leq 9$ ✓

b. $y \geq 9$

c. $y \leq 4$

d. All real numbers

The parabola opens downward because of the negative leading coefficient. The vertex is at $(4, 9)$, which is a maximum. Thus the range is $y \leq 9$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $y = f(x)$ is shifted 3 units to the right and 5 units down. What is the equation of the transformed graph?

a. $y = f(x - 3) - 5$ ✓

b. $y = f(x + 3) - 5$

c. $y = f(x - 3) + 5$

d. $y = f(x + 3) + 5$

A horizontal shift to the right by h replaces x with $x - h$. A downward shift of k subtracts k from the function. Thus $y = f(x - 3) - 5$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following functions is an even function, satisfying $f(-x) = f(x)$ for all real x?

a. $f(x) = x^3 - x$

b. $f(x) = 4x^2 + 7$ ✓

c. $f(x) = 2x + 1$

d. $f(x) = x^4 - 3x$

Check $f(x) = 4x^2 + 7$: $f(-x) = 4(-x)^2 + 7 = 4x^2 + 7 = f(x)$. Therefore, it is even.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the horizontal asymptote of the rational function $f(x) = \frac{6x^2 - 5x + 1}{2x^2 + 3x - 4}$?

a. $y = 0$

b. $y = 3$ ✓

c. $y = 2$

d. No horizontal asymptote

Since degrees of numerator and denominator are equal (both 2), the horizontal asymptote is the ratio of leading coefficients: $y = \frac{6}{2} = 3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A function is defined piecewise by $f(x) = 2x + 3$ for $x < 1$, and $f(x) = x^2 - 1$ for $x \geq 1$. What is the value of $f(-2) + f(3)$?

a. 7 ✓

b. 8

c. 9

d. 11

For $x = -2 < 1$: $f(-2) = 2(-2) + 3 = -1$. For $x = 3 \geq 1$: $f(3) = 3^2 - 1 = 8$. Sum $= -1 + 8 = 7$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $[x]$ denotes the greatest integer function (floor function), what is the value of $[3.8] - [-2.4]$?

a. 5

b. 6 ✓

c. 7

d. 1

$[3.8] = 3$ and $[-2.4] = -3$ (the greatest integer less than or equal to -2.4). Therefore, $3 - (-3) = 6$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of c is the piecewise function $f(x) = cx + 4$ for $x \leq 2$, and $f(x) = 3x^2 - 2$ for $x > 2$ continuous at $x = 2$?

a. 2

b. 3 ✓

c. 4

d. 5

For continuity at $x = 2$, the left limit must equal the right limit:

$c(2) + 4 = 3(2)^2 - 2 \Rightarrow 2c + 4 = 12 - 2 = 10 \Rightarrow 2c = 6 \Rightarrow c = 3$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the domain restriction for $f(x) = \frac{x+1}{x-9}$? (Give the excluded x-value.)

a. 11

b. 14

c. 3

d. **9** ✓

The denominator $x - 9$ is zero when $x = 9$; this value is excluded.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What value of x is excluded from the domain of $f(x) = \frac{1}{x}$?

a. -1

b. 1

c. 5

d. **0** ✓

The denominator is zero when $x = 0$, so $x = 0$ is excluded.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the standard (x, y) coordinate plane, what is the area, in square coordinate units, of the region enclosed by the graphs of the functions $f(x) = |x - 4| - 3$ and $g(x) = 5 - |x - 4|$?

a. 16

b. **32** ✓

c. 48

d. 64

To find the area of the enclosed region, first determine the points of intersection of the two graphs by setting $f(x) = g(x)$: $|x - 4| - 3 = 5 - |x - 4|$ Adding $|x - 4|$ and 3 to both sides yields: $2|x - 4| = 8 \Rightarrow |x - 4| = 4$ This gives two solutions for x: $x - 4 = 4 \Rightarrow x = 8$ $x - 4 = -4 \Rightarrow x = 0$ Evaluating the y-values at these points: - For $x = 0$: $y = |0 - 4| - 3 = 1$, giving the intersection point $(0, 1)$. - For $x = 8$: $y = |8 - 4| - 3 = 1$, giving the intersection.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For $y = (x + 3)^2 + 2$, what is the axis of symmetry?

a. x = 0

b. x = -5

c. **x = -3 ✓**

d. x = -1

The axis of symmetry of $a(x - h)^2 + k$ is $x = h = -3$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $f(x) = x - 3$, find the average rate of change from $x = 1$ to $x = 3$.

a. 3

b. -3

c. 5

d. **1 ✓**

$\frac{f(3) - f(1)}{3 - 1} = \frac{0 + 2}{2} = 1$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A cylinder has radius 5 and height 4. What is its volume (in terms of π)?

a. 144π

b. 100π ✓

c. 40π

d. 20π

Volume $= \pi r^2 h = \pi(5)^2(4) = 100\pi$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

A right triangle has hypotenuse 13 and one leg 5. What is the length of the other leg?

a. 28

b. 33

c. 22

d. **12 ✓**

$b = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = 12$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A right triangle has legs 3 and 4. What is the length of the hypotenuse?

a. 7

b. **5** ✓

c. 29

d. 13

$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5.$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A circle has radius 6. What is its area (in terms of π)?

a. 49π

b. 36π ✓

c. 12π

d. 72π

Area $= \pi r^2 = \pi(6)^2 = 36\pi.$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A circle has diameter 18. What is its area (in terms of π)?

a. 162π

b. 18π

c. 100π

d. 81π ✓

Area $= \pi r^2 = \pi(9)^2 = 81\pi.$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Coordinate Geometry

A rectangular prism has dimensions $6 \times 3 \times 6$. What is its volume?

a. 203

b. 61

c. 235

d. **108** ✓

Volume $= lwh = (6)(3)(6) = 108.$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Plane Geometry

An isosceles trapezoid has parallel bases of lengths 4 cm and 16 cm. A circle is inscribed inside the trapezoid such that it is tangent to all four sides. What is the area, in square centimeters, of the region inside the trapezoid that lies outside the circle?

a. $80 - 16\pi$ ✓

b. $80 - 25\pi$

c. $100 - 25\pi$

d. $64 - 16\pi$

1. **Determine the leg lengths of the trapezoid:** Let the trapezoid be $ABCD$ with parallel bases $AB = 4$ cm and $CD = 16$ cm, and nonparallel legs $AD = BC$. Because a circle is tangent to all four sides, the trapezoid is a tangential quadrilateral. Tangent segments drawn from any vertex to the circle are equal in length. By symmetry, the points of tangency on the bases bisect each base: - The segments from the top vertices to the top tangency point each measure $4/2 = 2$ cm. - The segments from the bottom ver.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

A box contains 10 cards, each labeled with a distinct integer from 1 through 10. If 3 cards are drawn at random from the box without replacement, what is the probability that the sum of the numbers on the 3 drawn cards is divisible by 3?

a. $\frac{1}{4}$

b. $\frac{3}{10}$

c. $\frac{1}{3}$

d. $\frac{7}{20}$ ✓

To find the probability that the sum of the 3 chosen integers is divisible by 3, we synthesize combinatorial counting with modular arithmetic. 1. **Total Number of Outcomes:** The total number of ways to choose 3 cards from 10 without replacement is given by the combination formula: $\binom{10}{3} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$ 2. **Partition by Remainder modulo 3:** Group the integers 1 through 10 by their remainder when divided by 3: - **Remainder 0:** $\{3, 6, 9\}$ (3 cards) - **Remain.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability

In the standard (x, y) coordinate plane, a triangle has vertices at $O(0, 0)$, $A(8, 0)$, and $B(2, 6)$. A horizontal line with equation $y = c$, where $0 < c < 6$, divides the interior of ΔOAB into two regions of equal area. What is the value of c?

a. $6 - 3\sqrt{2}$ ✓

b. 3

c. $3\sqrt{2}$

d. $6 - 2\sqrt{3}$

To find the value of c, we first determine the total area of ΔOAB. The base along the x-axis lies on the segment from $O(0, 0)$ to $A(8, 0)$, which has length $b = 8$. The height of the triangle corresponds to the y-coordinate of vertex $B(2, 6)$, so $h = 6$. $\text{Area}(\Delta OAB) = \frac{1}{2} \times 8 \times 6 = 24$ The horizontal line $y = c$ is parallel to the base $\overline{OA}$ (which lies on the line $y = 0$). A line parallel to one side of a triangle creates an upper triangle that is s.

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